

INTAKE-ARBITER -- FREE-COOLING DECISION REPORT

Data centres over-cool, continuously, because nobody can promise them the hours ahead. A chiller plant needs hours of notice to change mode, and a thermometer only ever reports NOW -- so the mechanical chillers keep running through hours that outside air could have cooled for free. FortyGuard closes that gap with heat intelligence 2 m above the ground, the height a ground-mounted condenser actually breathes. This agent turns that forecast into an hour-by-hour schedule carrying a calibrated safety bound.

THE SITE

Location Google, TX -- station KLNC, America/Chicago
Committed pair OSM 1426525217 -> 1426525219
Compass Data Center / Compass Data Center
Facade gap 199.5 m between the two halls
Weather record 42,375 real hourly records from KLNC
Plume physics 576 steady-state solves on this site's own footprints; worst intake rise 0.4946 C at 230 deg

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day 2022-01-29 (knife_edge)
Plant limit 18.0 C
Notice required 3 h
Forecast skill 0.50 relative to persistence
Level anchor sensor
Condenser bank longest facade
Switch budget 1 mode changes per day
Minimum dwell 3 h
Max dew point 15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 6 of 24 hours with 1 mode change. The reactive incumbent took 14 hours -- 8 more -- but declared 1 unsafe hour against the agent's 0. The incumbent had to break its own switch budget 1 time to stay safe; the agent never did. 13 hours were safe but still ran chillers, because the schedule could not afford them.

Agent 6 of 24 hours free cooling, 1 mode change(s), 0 unsafe hour(s)
Incumbent 14 of 24 hours free cooling, 2 mode change(s), 1 unsafe hour(s)
the incumbent broke its own switch budget 1 time(s) to stay safe
13 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	mechanical	yes	switch budget	0.394	18.0	0.245	1.074
01	mechanical	yes	switch budget	0.229	18.0	-0.365	1.056
02	mechanical	yes	switch budget	-0.136	18.0	-1.145	0.984
03	mechanical	yes	switch budget	-0.301	18.0	-1.535	0.949
04	mechanical	yes	switch budget	-0.466	18.0	-1.255	0.928

05	mechanical	yes	switch budget	-1.051	18.0	-1.755	0.930
06	mechanical	yes	switch budget	-1.471	18.0	-2.255	0.869
07	mechanical	yes	switch budget	-0.191	18.0	-0.975	0.960
08	mechanical	yes	switch budget	4.144	18.0	4.855	1.533
09	mechanical	yes	switch budget	8.077	18.0	10.161	2.300
10	mechanical	yes	switch budget	11.433	18.0	13.099	2.660
11	mechanical	yes	switch budget	14.483	18.0	15.463	2.025
12	mechanical	yes	switch budget	17.078	18.0	17.551	1.424
13	mechanical	no	dry-bulb	18.288	18.0	18.771	1.127
14	mechanical	no	dry-bulb	19.553	18.0	19.278	1.008
15	mechanical	no	dry-bulb	19.941	18.0	19.703	0.892
16	mechanical	no	dry-bulb	19.996	18.0	19.533	0.951
17	mechanical	no	dry-bulb	19.252	18.0	17.709	1.052
18	FREE	yes	-	17.400	18.0	15.330	1.245
19	FREE	yes	-	16.109	18.0	13.269	1.484
20	FREE	yes	-	13.872	18.0	11.879	1.673
21	FREE	yes	-	12.589	18.0	10.525	1.746
22	FREE	yes	-	10.688	18.0	8.918	1.462
23	FREE	yes	-	10.384	18.0	8.819	1.196

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin
actual = what the intake temperature turned out to be, from the station record
margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 0.394 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

01:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 0.229 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

02:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is -0.136 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

03:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is -0.301 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

04:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is -0.466 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a

thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

05:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is -1.051 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

06:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is -1.471 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

07:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is -0.191 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

08:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 4.144 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

09:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 8.077 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

10:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 11.433 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

11:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 14.483 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

12:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 17.079 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is

also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

13:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.288 C against a 18.0 C limit, so it fails by 0.288 C. It would take a limit 0.288 C higher, or a bound 0.288 C tighter, to change this hour.

14:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 19.553 C against a 18.0 C limit, so it fails by 1.553 C. It would take a limit 1.553 C higher, or a bound 1.553 C tighter, to change this hour.

15:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 19.941 C against a 18.0 C limit, so it fails by 1.941 C. It would take a limit 1.941 C higher, or a bound 1.941 C tighter, to change this hour.

16:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 19.996 C against a 18.0 C limit, so it fails by 1.996 C. It would take a limit 1.996 C higher, or a bound 1.996 C tighter, to change this hour.

17:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 19.252 C against a 18.0 C limit, so it fails by 1.252 C. It would take a limit 1.252 C higher, or a bound 1.252 C tighter, to change this hour.

18:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 17.400 C, which is 0.600 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.245 C, and the margin is measured, not chosen: 1.113 C of group-conditional forecast error for this hour of day, plus 0.1318 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

19:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 16.109 C, which is 1.891 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.484 C, and the margin is measured, not chosen: 1.351 C of group-conditional forecast error for this hour of day, plus 0.1334 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

20:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 13.872 C, which is 4.128 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.673 C, and the margin is measured, not chosen: 1.576 C of group-conditional forecast error for this hour of day, plus 0.0974 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

21:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 12.589 C, which is 5.411 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.746 C, and the margin is measured, not chosen: 1.597 C of group-conditional forecast error for this hour of day, plus 0.1487 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

22:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 10.688 C, which is 7.312 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.462 C, and the margin is measured, not chosen: 1.346 C of group-conditional forecast error for this hour of day, plus 0.1158 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

23:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 10.384 C, which is 7.616 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.196 C, and the margin is measured, not chosen: 1.063 C of group-conditional forecast error for

this hour of day, plus 0.1334 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.
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